

2013–2014 Topology and control

SPB introduction and dedicated control and modulation [36,42].



2015–2017 Stability and architectural assessment

Dc-link stability and balancing requirements [44,45]; comparison with PPB-I/II and MHF candidates [48].



2017–2018 Modulation and distributed control

Switching-loss and dc-link harmonic shaping [46]; distributed balancing [47].



2019–2021 Converter–machine coordination

Interleaving, dc-link ripple, and machine phase-count selection [33, 49–51].



2023–2024 Reconfiguration and device integration

Common-duty-ratio and decentralized balancing, power sharing, and gate-delay effects [52–55]; online module reconfiguration [56]; 650-V GaN SPB comparison with a three-level flying-capacitor converter [57].



Open research direction

Unified SPB design for EV applications